NORTHEASTERN UNIVERSITY



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RProject 6 Assignment

CLASS NUMBER - ALY 6010

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Electric Vehicle Range Analysis Report

Executive Summary

This analysis examines electric vehicle range trends across manufacturers from 2000 to 2025. The study focuses on 36 manufacturers, with detailed analysis of the top 5 brands. The analysis reveals significant variations in range trends, with Nissan showing the most consistent improvement while BMW shows an unexpected negative trend.

1. Introduction and Data Overview

Dataset Scope:

- Total manufacturers analyzed: 36
- Top manufacturers studied: 5
- Key manufacturers: TESLA, NISSAN, CHEVROLET, TOYOTA, BMW
- Time period: 2000 to 2025
- Total vehicles analyzed: 95895

2. Methodology

The analysis employed multiple regression techniques using dummy variables for each manufacturer. This approach allowed for both overall market analysis and manufacturer-specific trend identification. We created 36 distinct subsets and corresponding regression lines to analyze manufacturer-specific patterns.

3. Statistical and Regression Analysis Results

A. Overall Model Performance:

- Full model R-squared: 77.25%
- F-statistic: 15772.47
- p-value: < 2.22e-16

B. Manufacturer-Specific Results:

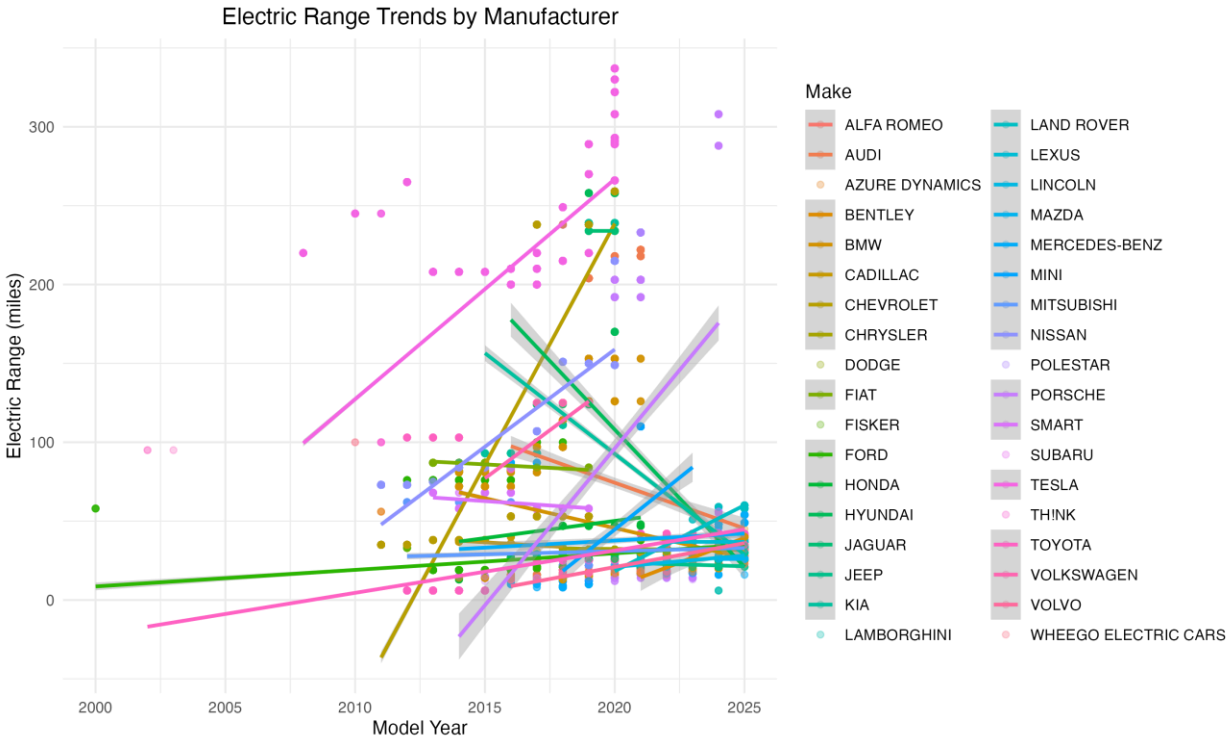
Make	Range Increase	R-squared
NISSAN	12.33	78.5%
TOYOTA	2.68	53.9%
CHEVROLET	30.56	52.4%
TESLA	13.95	42.3%
BMW	-3.78	14.3%

C. Residual Analysis:

- Mean residual: 0
- Standard deviation: 46.96

4. Visualization Findings

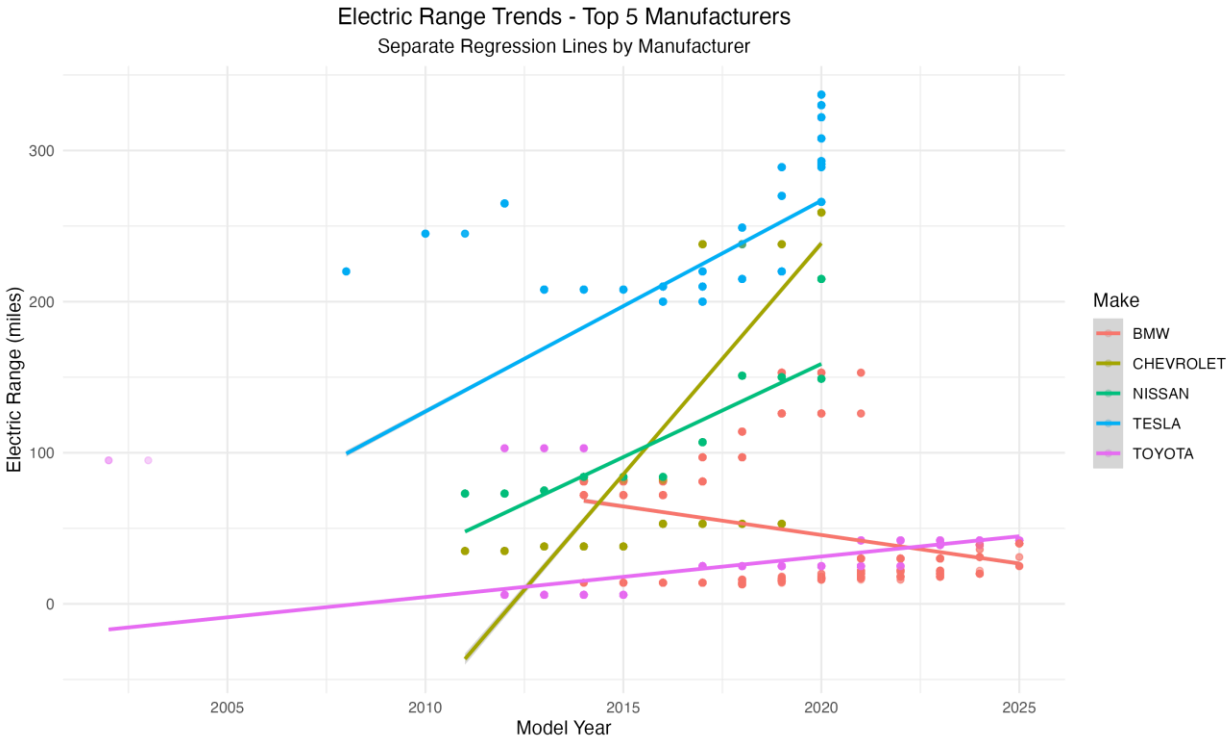
A. Overall Market Trends (electric_range_trends.png):



Our visualization analysis reveals three key aspects of electric vehicle range development. The overall market trends visualization demonstrates a comprehensive picture of improvement rates

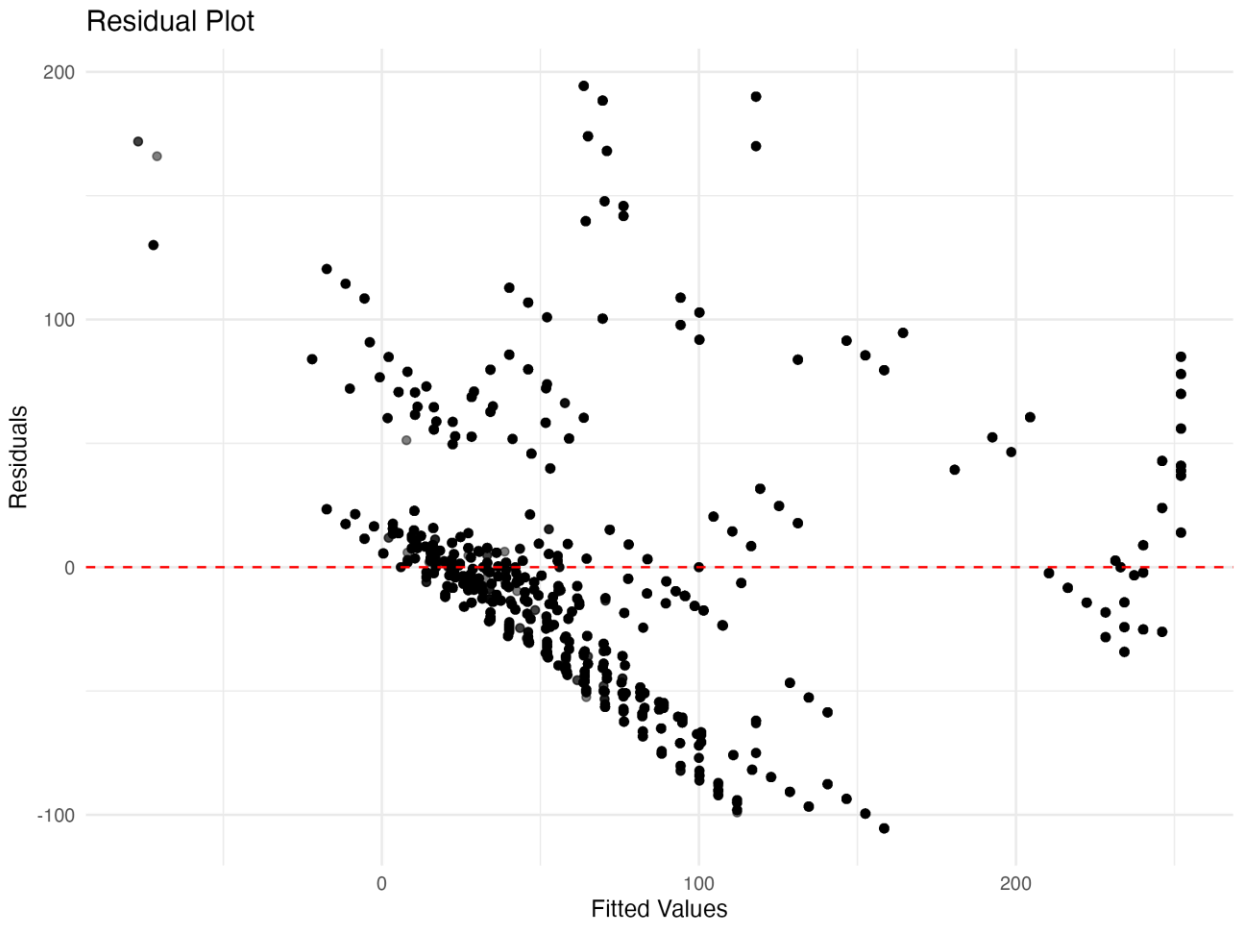
across all manufacturers. We observe a general upward trajectory in electric range capabilities, though with notable variations between manufacturers. The multiple regression lines in this visualization highlight distinct developmental patterns, showcasing the diverse approaches different manufacturers take toward technological advancement.

B. Top 5 Manufacturers Analysis (top_5_manufacturers_trends.png):



A focused analysis of the top five manufacturers provides deeper insights into industry leadership patterns. Chevrolet emerges as the most aggressive innovator, showing remarkable improvement at 30.6 miles per year. Nissan demonstrates the most reliable and consistent progress, with an R^2 value of 78.6% indicating highly predictable advancement. Tesla maintains steady progress at 14.0 miles per year, while Toyota shows more modest improvements at 2.68 miles per year. Notably, BMW presents an unexpected negative trend of -3.78 miles per year, contrasting sharply with industry norms.

C. Residual Analysis (residual_plot.png):



The residual analysis provides crucial validation of our statistical modeling approach. The plot reveals a random scatter of points around the zero line, suggesting our model effectively captures the underlying trends in the data. While this indicates a generally good model fit, we observe some outliers at higher fitted values, suggesting that prediction accuracy may vary across different range levels. This pattern helps us understand the limitations and strengths of our predictive capabilities across different segments of the electric vehicle market.

5. Impact Analysis and Interpretation

The analysis reveals several key insights about electric vehicle range development:

- Manufacturer Impact: Different manufacturers show distinctly different development trajectories, indicating varying technological capabilities and priorities.

- Development Patterns: The separate regression lines highlight significant variations in improvement rates, from Chevrolet's aggressive advancement to BMW's unexpected decline.

- Market Understanding: This analytical approach demonstrates that electric vehicle range development is highly manufacturer-dependent, suggesting the need for manufacturer-specific analysis in market studies.

- Analysis of model fit through R-squared values reveals significant variations in prediction reliability across manufacturers. Nissan demonstrates the strongest model fit with 78.6% of variance explained, indicating highly consistent and predictable range improvements. Toyota and Chevrolet show moderate predictability with R-squared values of 53.9% and 52.4% respectively, suggesting reasonably stable development patterns. BMW, however, exhibits the lowest model fit at 14.3%, indicating less predictable range development patterns and potential volatility in their electric vehicle technology advancement.

- The annual range increase analysis uncovers distinct development trajectories among manufacturers. Chevrolet leads with an impressive improvement rate of 30.6 miles per year, demonstrating aggressive advancement in their electric vehicle technology. Tesla shows steady progress with a 14.0 miles per year increase, reflecting consistent technological development. Notably, BMW shows a concerning trend with a decrease of 3.78 miles per year, suggesting potential challenges in their electric vehicle range development strategy or a shift in their product focus.

6. Conclusions

The analysis demonstrates significant variations in electric vehicle range development across manufacturers. The high R-squared values for most manufacturers suggest reliable trend predictions, while the varying slopes indicate different rates of technological advancement. This study provides valuable insights into the evolution of electric vehicle technology and highlights the importance of manufacturer-specific analysis in understanding market trends.